

17 – Solution Architecture

Stage 1 foundation artefact: module boundaries, event stream and queue, edge architecture, integration surface. Version 0.2 realigns the module map to nine phase-1 modules with Trading Documents as its own module (M4), and reconciles the architecture with documents 14, 18, 19, 20, 30 and 31.

Field	Value
Document	17 – Solution Architecture
Version / status	v0.2.1 – Draft for design review (supersedes v0.1 of 3 September 2026; v0.2.1 restores detail items dropped in the v0.2 merge)
Date	7 September 2026
Baseline it designs against	Requirements package v0.2 plus the v0.3 change requests of 3 September 2026, and the Stage 1 companions as drafted: 14 Technical Foundations v0.2, 18 Core Data Model v0.1, 19 Cross-Cutting Services v0.1, 20 Stage 2 Module Design Plan v0.2, 30 Frontend Foundations v0.1, 31 Application Update and Distribution v0.1
Audience	Design leads, technical leads, the DevOps engineer, and Federation admins for the business-visible decisions (sections 4, 6, 9.2, 10.3)
What changed	Section 0 lists every alignment change from v0.1 with its reason; ADR-28 is revised, ADR-37 to ADR-40 are new

How to read this document. Plain language for a mixed audience; abbreviations expanded on first use and collected in Appendix A. Each numbered decision is marked **Decided** (follows from an accepted requirement or decision record) or **Proposed** (needs review). Nothing here changes a requirement; where the architecture depends on an open item, the dependency is stated.

0. Alignment changes from v0.1

v0.1 was written before the core data model, cross-cutting services and Stage 2 plan existed. Those documents made choices v0.1 had left open or made differently. This version adopts them; each change is listed so the review can accept or reverse it deliberately.

#	Change	v0.1 position	v0.2 position	Reason
C1	Module count and grouping	Fourteen business modules plus kernel	Nine phase-1 modules (M1–M9), Procurement as M10 (phase 2), plus kernel	Doc 20 blueprints and handover to full-stack developers by module; fewer, thicker modules with clear contracts. Internal boundaries survive as packages.
C2	Trading documents	Split across Orders, Delivery & Receiving, Documents & Payments	One module, M4 Trading Documents: order, delivery note, GRN, discrepancy, claim, invoice, credit/debit note, payment receipt, statement, exposure	Order-to-settlement and the ownership pivot (ADR-03) belong together; the state machines are designed in one place (doc 20 DR-1).
C3	Party and security	Party & Tenancy and Governance & Security as two modules	M1 Party, Tenancy & Security	Everything scopes on both; one contract to freeze first.
C4	Batch ownership	Batch owned by Inventory	Batch identity (SKU, batch number, expiry, printed MRP, supplier) in M2; stock lot (holding, cost) and movements in M5	Doc 18: one physical batch is owned successively by several entities at different costs; identity and holding are different things.
C5	Repack recipe	Owned by Catalogue	Owned by M5	Execution, yield variance and reversal are inventory operations.
C6	Loss & Claims	Own module	Write-off and loss categories in M5; claims in M4	A claim is a trading-document lifecycle ending in a credit note; a write-off is a stock lifecycle.
C7	Localisation module	Business module owning translation content	Dissolved: interface catalogue and rendering are kernel (doc 19 section 5); catalogue names are M2; shop language is M1 location	No table or lifecycle was left for a module to own.
C8	Documents and numbering	Numbering a kernel service; document base implicit	Kernel owns the document base type (header, lines, links, state history, attachments) and numbering series; modules own document types	Doc 18 part C; till receipts, GRNs and back-office documents share one base.
C9	Event naming	PascalCase (SaleRecorded)	Dotted, versioned (receipt.issued.v1)	Doc 19 envelope; the version travels in the type.
C10	Update distribution	POS application's own update agent; LAN sharing between tills	Device-owner management agent (doc 31); no till-to-till sharing in phase 1	Google developer-verification exemptions require the device-owner path; single-till shops make LAN sharing marginal. NFR-UPD-040 needs a change request (section 15).
C11	Scope enforcement	Kernel scoping service first, row-level security second	Row-level security policy classes are the data-layer enforcement (doc 18 section 3.7); the kernel sets the session scope	One mechanism keyed on one column; the service layer checks permissions, the database checks scope.
C12	Reporting store	Replica plus OpenSearch in P2	Replica in phase 1; columnar store on trigger (doc 19 DR-3); analyst SQL role on the replica	Doc 20 Appendix A.
C13	Deployment	Web/API and worker profiles	Same, plus stateless scale-out rules and an ingest-only worker role for bursts	Doc 14 section 1.3.
C14	Product images	Not mentioned	M2 owns images; tills cache thumbnails	September 2026 decision.

1. Purpose, scope and what this document decides

This is the first design artefact of Stage 1. It fixes the shape of the system at the level everything else depends on: which parts exist, what each owns, how they may depend on one another, how business events flow, and where the outside world connects. It stops short of table-level detail (doc 18), the internals of the cross-cutting services (doc 19), provider-specific infrastructure (doc 14, doc 35) and screens (doc 30).

This document decides	This document defers
The architectural style: modular monolith with a message queue, plus an offline-first edge application	Provider choice, sizing, hosting (doc 14, doc 35)
The module map: nine phase-1 modules, Procurement as phase 2, a platform kernel; layers, dependency rules and their enforcement in the build	Table-level schema, indexes, partitions (doc 18)
Ownership of every document lifecycle, ledger and reference set	Internals of identity, permissions, audit, i18n (doc 19)
The event and queue design; edge ingestion; reference distribution to tills	Sync message formats and timing (doc 32)
The integration surface and the edge application's internal structure	Module designs (docs 21–29), report catalogue (workshop), screens (doc 30 and module designs), numbering's final form pending the tax advisor (NFR-CMP-030)

2. Architecture drivers

Driver	Source	Architectural response
A federation of independent legal entities; a shop is a location, not an entity	ADR-02, FR-PTY-010	One platform with entity-scoped tenancy: every record carries its owning entity (and location where relevant); visibility enforced at the data layer by row-level security, never in screens
Tills trade one to two days offline; no merge logic; one writer per record type	ADR-10, NFR-AVL-020, NFR-OFF-010	Edge application with its own database and gapless counters; strict split into till-owned (uploaded as events) and central-owned (distributed read-only)
Documents are immutable, corrected by linked compensating documents	NFR-AUD-010, doc 03	One document base type in the kernel; append-only; typed correction links; no update path for issued documents anywhere
Gapless multi-series numbering with no central coordination offline	ADR-11, FR-BIL-050/060, NFR-OFF-030	Numbering as a kernel service: entity, location and till-position series; locally persisted counters at the edge; gap detection on ingestion
10,000 shops, 15,000 tills, 3 million receipts a day; reconnection bursts	NFR-PRF-010/030, ADR-22	Queue-based idempotent ingestion with backpressure; small write set at central; partitioned ledgers; read models on a replica
One DevOps engineer	NFR-UPD-050, ADR-20	One deployable in two profiles; one database; standard components; no microservices, no orchestration platform in phase 1
Hosting not bound to a provider; local hosting possible	Q-01 change request	Containers on commodity open-source components; nothing proprietary in the critical path (doc 14)
Sinhala, Tamil, English with correct collation and thermal rendering	NFR-LOC, FR-LOC	ICU kernel; translations as data; raster receipts
Money recorded, never moved; no card data	ADR-01, ADR-19	No payment, bank or card-terminal integration
No statutory accounting; journal mapping per document type	ADR-05, FR-BIL-020	Integration surface as adapters; postings derived from documents
Phased delivery with a pilot gate	ADR-23	Phase 2 and 3 modules placed on the map now
Single-device, single-person shops	v0.3 change request	Shop-management mode in the POS application; web back office on the same terminal for online-only functions
Gazetted ceilings are law; MRP is a batch attribute; ceilings are date-effective	FR-PRC-040, ADR-08, control-price table	Pricing owns the control-price record; one shared price-resolution engine used identically at central and at the till
Full-stack developers take a module each	September 2026 decision	Modules are vertical slices with contracts and screen inventories (doc 20, doc 30); two shells owned by a platform pair

3. Architecture principles

#	Principle	In practice
P1	One system, many modules, explicit boundaries	One backend application; module boundaries enforced by the build (section 6), not by convention.
P2	One database, one schema per module, no cross-schema queries	A module owns its tables in its own PostgreSQL schema; it reaches another module only through its published interface or its events.
P3	State changes cross module boundaries only as events	Synchronous calls only downward to master-data modules and for read-only queries; changing another module's state is a published event handled asynchronously.
P4	Documents are facts; facts are never edited	Issued documents are immutable; corrections are new linked documents, at the till and in the back office alike.
P5	Exactly one writer per record type	The till writes its own sale, session, receiving and count records; central writes everything else; nothing is written in both places.
P6	Scope is enforced at the data layer, not in screens	Row-level security keyed on the owning entity and location; the kernel sets the session scope from the token; a screen cannot leak what the database will not return.
P7	The same business rule runs the same way everywhere	Price resolution, rule evaluation, tax, rounding, envelope and counters are one shared library used by central and the Android application.
P8	Provider-neutral by construction	Only components that run on any cloud or on premises are hard dependencies.
P9	Everything privileged is audited, append-only	Audit is a kernel service with a write-only path; no module can bypass it; no role can alter it.
P10	Configuration over code, within limits	What the baseline marks configurable is data with a default; what is law (gazetted ceilings, segregation of duties) is code.
P11	Extractable later, monolithic now	Any module could be lifted into a service if load and team justify it; the option is preserved, not exercised.
P12	Stateless application tier	Any request to any instance; no sticky sessions; idempotent commands; singleton jobs under cluster locks (doc 14 section 1.3).

4. System context

The system sits between the four tiers of the federation and a small set of external systems. Two boundaries are easy to assume the other way: card terminals are not connected (the cashier keys a reference), and the societies' accounting systems are not connected in phase 1 (an export is produced later).

Actor or external system	Relationship to the system
Federation (COOPFED) staff	Steward the shared catalogue and images, run procurement (phase 2), sell to distributors, record payment receipts, keep the gazetted control-price register, view every entity's data read-only, administer their own users only
Distributor staff (about 75)	Buy from the Federation, sell to their MPCS on negotiated price lists, may run shops
MPCS staff (about 800)	Buy from a distributor, set one retail price list for all their shops, author discount rules, approve adjustments and write-offs, administer shop users and roles
Shop staff (about 4,000 shops, growing to 10,000)	Sell, receive, count, request write-offs, offline-capable; a single-staff shop does all of it from the terminal
Drivers	Proof of delivery on a mobile app or paper; vehicle operations out of scope
Auditors; Department of Co-operative Development	Read-only: entity-scoped for society auditors, federation-wide for Federation auditors, time-bound for the regulator
Helpdesk and operations bridge	Sync health, fleet and queue monitoring, diagnostics; no transactional access; cannot reset MPCS passwords
Accounting systems (phase 3)	Receive journal-posting exports; the first API consumer
Inland Revenue e-invoicing (when mandated)	Reserved adapter at invoice issuance
SMS / e-mail gateway	Notifications through a swappable adapter
Consumer Affairs Authority gazettes	Loaded manually by a named Federation role as date-effective control prices; no automated feed exists
Device management (doc 31)	Provisioning, silent updates, lock and wipe of terminals through a device-owner agent

5. Container view

The running parts. The two carrying business logic (central application, POS edge application) are what the module map describes; the rest is infrastructure.

Container	Technology	Responsibility
Central application	Java 21 / Spring Boot 3 modular monolith; Spring Modulith boundary enforcement (ADR-32)	All modules and the kernel in one artefact. Web/API profile serves the back-office web, edge sync API, driver API and third-party API; worker profile runs ingestion (with an ingest-only role for bursts), event consumers, rendering, notifications and scheduled jobs. Stateless; scales by adding instances (P12).
PostgreSQL	16 or later; managed or self-hosted per doc 14; connection pooler in front	Single system of record: one cluster, one schema per module plus kernel schemas (documents, numbering, events, audit). Primary with streaming replica for reporting and the analyst role.
Message broker	RabbitMQ quorum queues (proposed default); NATS JetStream alternative; not Kafka	Work-queue delivery and fan-out. Not the system of record for events: the outbox archive in PostgreSQL is.
Object storage	Any S3-compatible store; MinIO on premises	Photographs, product images, rendered A4 output, signed packages, exports, audit hash anchors (write-once bucket).
Back-office web	React + TypeScript single-page application (doc 30)	All back-office functions, role-scoped; responsive so a terminal's browser can open it for online-only shop functions.
POS edge application	Native Android (Kotlin, Compose) on ARM terminals; SQLite via Room; device-owner kiosk (doc 30, doc 31)	Selling, receiving, counting, write-off drafts, cash management, reports, shop-management mode, sync worker, peripheral layer, receipt renderer. Offline-capable one to two days.
Driver application	Same Android stack, minimal	Drop list and proof of delivery.
Identity provider	Self-hosted OIDC provider, single realm (doc 19)	Login, tokens, TOTP, device credentials.
Device management agent and update service	Device-owner MDM class plus a small update service (doc 31)	Provisioning, silent install, rings, health gates, version floor, lock and wipe.
Edge, proxy, observability	CDN/WAF edge; reverse proxy; Prometheus/Grafana/Loki class	TLS, routing, rate limiting of the sync API; metrics, logs, alerts.

6. Module map, layers and dependency rules

Four layers. A module may call downward and query sideways read-only; it never calls upward and never changes another module's state except by an event the other module chooses to handle. The graph is acyclic.

Layer	Modules	Role and rule
0 – Platform kernel	Scope and session; Documents & Numbering base; Audit; Event backbone (outbox, relay, inbox); Edge sync gateway; Internationalisation; Attachments; Notifications delivery; Configuration register; Scheduling	Technical services with no business rules; every module may use them; they depend on nothing above (doc 18 parts C and D, doc 19).
1 – Master data and policy	M1 Party, Tenancy & Security; M2 Catalogue & Batch; M3 Pricing & Rules	Slowly changing reference data and the policies governing transactions. Their published interfaces are what transaction modules call synchronously; their changes reach tills as reference-data snapshots (8.5).
2 – Transactions	M4 Trading Documents; M5 Inventory, Costing, Repack & Loss; M6 POS & Offline; M7 Customers & Payment Recording; M10 Procurement (phase 2)	Documents and ledgers. Each owns document lifecycles from doc 03. They interact only by events and read-only queries.
3 – Read side	M8 Reporting Pipeline; M9 Integration	Consume the event stream to build read models, reports, exports and outbound adapters; write nothing to layers 1 and 2.

6.1 Dependency rules (enforced in the build)

#	Rule	Enforcement
R1	A module exposes exactly one published package: commands, queries, event types, value types. Everything else is internal.	Spring Modulith verification tests (or ArchUnit); the build fails on violation.

#	Rule	Enforcement
R2	Layer 2 modules may call layer 1 published APIs synchronously; layer 2 modules call each other only through read-only query interfaces marked as such.	Query interfaces are a distinct annotated type.
R3	Any change to another module's state is a domain event handled asynchronously; no cross-module command in one transaction.	Command interfaces are not exported across layer 2.
R4	A module owns its PostgreSQL schema; no SQL crosses a schema; no cross-schema foreign keys; references by identifier.	Database roles per schema; migrations scoped per module; schema-ownership test.
R5	Kernel services never import business modules.	Package rules as R1.
R6	Reporting reads events and its own read models; it may query layer 1 for labels at render time.	Package rules as R1.
R7	Every mutating command carries an idempotency key; every singleton job runs under a cluster lock.	Framework interceptor; scheduler configuration test (P12).

Decision (Proposed – ADR-28, revised). Nine phase-1 modules, Procurement as a phase-2 module with its boundary present from day one, and a platform kernel, in four layers, with rules R1–R7 enforced by automated build checks from the first sprint.

7. Module catalogue

One entry per module: what it owns (the only module that writes those records), the SRS area it implements, its lifecycles, the events it publishes and consumes, and its phase. Event names use the doc 19 form; the working set is refined in the module designs, the ownership is the decision here. Doc 20 carries each module's commands, queries, deliberate omissions and risks; they are not repeated.

7.1 M1 – Party, Tenancy & Security

Field	Content
SRS area	FR-PTY; FR-SEC; FR-GOV
Owns	Entity with tier and VAT registration; location with owner, language, trading hours, size band, primary till; till position; device; trading relationship (price-list reference, credit limit, terms, dispute window, lock window, allocation rule; effective-dated); user, permission catalogue, role builder and templates (including shop-in-charge), scoped role assignment, external time-boxed grants, segregation-of-duties pairs and modes; session and idle-lock policy; sandbox/training flag per user; financial-year settings; visibility policy as data for the row-level security classes
Documents / lifecycles	None (master data); relationship activation and credit-limit change are permission-gated, audited acts
Publishes	entity.*, location.*, till_position.*, device.*, relationship.opened/amended.v1, user.*, role.changed.v1, external_grant.*, authentication.failed.v1, account.locked_out.v1
Consumes	None
Notes	The relationship is the anchor for price lists and credit (ADR-04) and is not a hard-coded tier, so Federation-to-MPCS direct trading later is configuration. Authentication is the kernel's identity provider (doc 19); authorisation policy lives here and is evaluated by the kernel on every request.
Phase	P1

7.2 M2 – Catalogue & Batch

Field	Content
SRS area	FR-SKU; FR-LOC-020 (trilingual master data)
Owns	SKU (global namespace, owner, status draft/local/shared, mandatory English name, trilingual names, bounded attributes); units and conversions; barcodes (many-to-one, optional batch link, symbology); tags; tax categories and rates; images (per SKU, optional per barcode); alias mapping on promotion; batch as physical identity (SKU, batch number, supplier, manufacture and expiry, printed MRP); synthetic batches for non-batch-tracked SKUs; a minimal supplier table until M10
Documents / lifecycles	SKU status lifecycle; promotion with alias; batch correction (new batch, audited)
Publishes	sku.*, barcode.registered/retired.v1, conversion.defined.v1, tag.defined.v1, tax_rate.published.v1, image.attached/retired.v1, batch.registered/corrected.v1
Consumes	M4 grn.confirmed.v1 and M5 repack.executed.v1 command batch registration
Notes	Change C4: batch identity is here, holding and cost are M5. The catalogue is the largest reference set distributed to tills; thumbnails ride in the snapshot for each shop's assortment. The steward's tooling (bulk import, duplicate detection across scripts, translation completeness) is first-class (T-03).
Phase	P1

7.3 M3 – Pricing & Rules

Field	Content
SRS area	FR-PRC; G section; F-07
Owns	Trade price lists per relationship (uniform Federation list; negotiated, volume-tiered distributor lists); one retail list per MPCS; Federation advisory prices; discount and markdown rules (closed vocabulary; the sole source of any reduction, ADR-14); gazetted control price (date-effective, SKU or tag, gazette reference; hard block); multi-MRP resolution policy per SKU or tag; cash rounding; per-customer preferential price (phase 3)
Documents / lifecycles	Rule lifecycle (draft, active, expired, withdrawn); price-list publication with apply_from
Publishes	price_list.published.v1, discount_rule.activated/retired.v1, control_price.entered/rescinded.v1, mrp_policy.changed.v1
Consumes	batch.registered.v1 (raise a pricing review when an incoming batch MRP is below the standing list)

Field	Content
Notes	Owns the price-resolution engine: a pure library with no database access, packaged once, used by M4 (inter-entity), by ingestion re-validation and by the till offline (P7, ADR-31). Resolution is min(list, lowest in-stock batch MRP, control price in force), then rules, then the control price re-checked; the receipt line records what bound it (doc 18 section 4.9).
Phase	P1 (preferential pricing P3)

7.4 M4 – Trading Documents

Field	Content
SRS area	FR-ORD; FR-DEL; FR-BIL (except numbering internals); A, C, D, I-05 to I-07, J sections
Owns	Order (buyer's series) with lines, allocation runs, backorders, lock window, indicative ETA; delivery note with drops (per ship-to location, with a bill-to entity for direct-to-shop deliveries), vehicle, driver, timestamps; proof of delivery; GRN (location series when captured at a shop; offline-capable; the ownership pivot); discrepancy (automatic on mismatch) and claim (raise, inspect, approve, reject); invoice, credit and debit notes; payment receipt with settlement to specific invoices and cheque sub-lifecycle; statements; committed exposure per relationship (warn-only); journal-posting mapping per document type; immutable invoice snapshot for e-invoicing readiness; intra-MPCS lateral transfer request (mediated)
Documents / lifecycles	Order, delivery note, GRN, discrepancy, claim, invoice, credit/debit note, payment receipt (doc 03 sections 2–8); the six state machines designed in doc 24; numbering addendum 24A after the tax advisor
Publishes	order.*, delivery_note.*, grn.captured/confirmed/closed.v1, discrepancy.*, claim.*, invoice.issued.v1, credit_note.issued.v1, debit_note.issued.v1, payment_receipt.recorded/disputed/reversed.v1, cheque.cleared/bounced.v1, exposure.updated.v1, statement.generated.v1, journal.postings_ready.v1
Consumes	M7 customer payments where the payer is a customer; M10 purchase orders for expected receipts (P2)
Notes	grn.confirmed.v1 is the most important event in the system: M2 registers batches from it, M5 creates the receiving entity's lots and posts receipts, invoicing is permitted by it, and loss attribution depends on its timestamp. Till-captured GRNs are ingested through the same path as receipts. Money is never moved (ADR-01); exposure warns, never blocks (ADR-12).
Phase	P1 (e-invoicing live P3; reorder suggestions feeding orders P2)

7.5 M5 – Inventory, Costing, Repack & Loss

Field	Content
SRS area	FR-INV; FR-LOSS; E, F sections; B-06; R-02
Owns	Stock lot (entity holding of a batch at a location, acquisition cost); stock movement ledger (append-only, every movement cites a document); derived balances; weighted-average cost per entity; FEFO suggestion; intra-entity transfer at cost; count tasks and results; adjustment with value-threshold approval; repack recipe and execution with expected yield and reversal; write-off with loss categories, requester ≠ approver, witness (remote for single-staff shops), photographs, value-banded approval; loss tolerances; expiry warning tasks; negative-stock and oversell exception records; opening balances; ageing and reorder points (phase 2)
Documents / lifecycles	Transfer, adjustment, repack, write-off, count, opening balance (doc 03 sections 9–11)
Publishes	stock.moved.v1, lot.negative.v1, count.*, adjustment.approved.v1, writeoff.requested/witnessed/approved/posted.v1, repack.executed/reversed.v1, batch.expiring.v1, opening_balance.recorded.v1
Consumes	grn.confirmed.v1 (lots and receipts), receipt.issued/refunded/voided.v1 via ingestion (deductions and returns per batch), claim.approved.v1 (return movements), landed_cost.finalised.v1 (P2 restates lot cost)
Notes	Provides the availability query for M4 and the lot list distributed to tills (batch, MRP, expiry, FEFO order). Concurrent offline oversells are accepted and flagged (oversell exception), never rejected. Changes C4–C6 apply.
Phase	P1 (ageing, reorder points, ABC cycle counting P2)

7.6 M6 – POS & Offline

Field	Content
SRS area	FR-POS; FR-PAY (till subset); FR-REP-030; H, I-03, O-02, P-05, Q-07 sections
Owns	On the edge: till session (float, drops, blind close, variance); receipt with lines, batch attribution, applied rules, tax, tenders, operator; void and same-session refund; parked sales; receipt rendering; local reference cache and version; local counters for the till-position series and, on the primary till, the location series; local outbox and upload cursor; shop-management mode hosting M4 and M5 screens (GRN, count, write-off draft, shop repack); training namespace. At central: ingestion – validation, idempotent acceptance, gap detection, re-validation of price and ceiling compliance with the shared engine, re-publication into the backbone; sync-health view per till
Documents / lifecycles	Receipt, refund, void, till session (doc 03 section 13)

Field	Content
Publishes	receipt.issued/voided/refunded.v1, till_session.opened/closed.v1, cash.dropped.v1, drawer.opened_no_sale.v1, mrp_cap.applied.v1, mode.switched.v1, sync.batch_received/anomaly.v1, till.heartbeat.v1
Consumes	Reference-data snapshots from layer 1 and M5 through the distribution channel (8.5), and the permission snapshot from M1; not domain events
Notes	Section 9. Ingestion never rejects a well-formed event for business reasons: a bad price is flagged, not refused; the sale already happened. Card, wallet and QR tenders are references only (ADR-19). Updates arrive through the management agent (doc 31), not through this module.
Phase	P1 (LAN peer stock visibility P2, NFR-OFF-040)

7.7 M7 – Customers & Payment Recording

Field	Content
SRS area	FR-CUS; FR-PAY (customer subset); I-01, I-02, I-04; K section
Owns	Registered customer (internal id, name, optional NIC where credit is granted, bounded attributes); phone numbers with audited change and reuse detection; account per MPCS (limit, balance, ageing, offline cap); account postings (credit sales, repayments, adjustments); customer payment receipts settling specific charges; banking record of takings
Documents / lifecycles	Account posting (append-only), customer payment receipt, opening balance from migration
Publishes	customer.registered/phone_changed.v1, account.opened/limit_amended/closed.v1, account.charged/credited.v1, banking.recorded.v1
Consumes	receipt.issued/refunded.v1 where a tender is a customer account; payment_receipt.recorded.v1 where the payer is a customer
Notes	Unrelated to cooperative membership (ADR-16). Tagged and credit customers with balances are part of a shop's snapshot so account sales work offline against the last known balance up to a configurable cap; minimal PII (ADR-27).
Phase	P1 (preferential pricing P3)

7.8 M8 – Reporting Pipeline

Field	Content
SRS area	FR-REP; O section; M-04, M-06; N-06
Owns	Read models in a reporting schema (sales by period, entity, shop, SKU, rule; stock position and ageing; discrepancy and claim register; loss by category; discount cost by rule; VAT summary; exposure and statements; till variance; exception queues); per-entity and per-location freshness stamp; report catalogue definitions; end-of-day aggregates; the A4 rendering pipeline; spreadsheet export; the analyst SQL role on the replica
Documents / lifecycles	None; read-only
Publishes	report.rendered.v1, export.ready.v1, end_of_day.completed.v1, projection.rebuilt.v1, exception_report.generated.v1
Consumes	Every domain event
Notes	Every federation-level figure carries the freshness of its least-recent contributor (FR-REP-020). Read models are rebuildable from the outbox archive, which is what makes a fixed catalogue safe to extend after the workshop. Heavy reads run on the replica; a columnar store follows on trigger (doc 19 DR-3).
Phase	P1 catalogue; self-service builder backlog

7.9 M9 – Integration

Field	Content
SRS area	FR-BIL-110, FR-BIL-140; R-05 to R-09; J-02, J-04
Owns	API clients and scoped grants; the OpenAPI surface as the versioned public contract; journal export using M4's posting maps; e-invoice submission adapter stub; notification templates and the provider adapter; webhooks (phase 2/3)
Documents / lifecycles	None
Publishes	notification.sent/failed.v1, journal.exported.v1, api_client.registered.v1
Consumes	journal.postings_ready.v1, invoice.issued.v1 (e-invoice seam), delivery and statement events for notifications
Notes	REST only; an adapter is a translator, never a place for business rules (anti-corruption layer). The first API consumer is the accounting system.

Field	Content
Phase	P1 (accounting integration and e-invoicing P3)

7.10 M10 – Procurement (phase 2)

Field	Content
SRS area	FR-PRO; L section
Owns	Supplier master (takes over M2's minimal table); supplier contracts; quotation register with audited award; purchase order; market price observations; shipment and landed cost; supplier performance; demand-forecast inputs
Documents / lifecycles	Purchase order, quotation register, shipment
Publishes	purchase_order.issued.v1, quotation.awarded.v1, landed_cost.finalised.v1, supplier_performance.computed.v1
Consumes	grn.confirmed.v1 at Federation warehouses; sales aggregates via M8 for forecasting
Notes	Placed on the map now so Federation warehouse receiving in phase 1 can carry a purchase-order reference field. Federation repacking (P3) reuses M5's recipe mechanism by permission.
Phase	P2 (repacking P3)

7.11 Platform kernel services

Service	Responsibility	Designed in
Scope and session	Sets the caller's entity and location scope and policy class on every database session from the token; row-level security policies enforce it	Doc 18 section 3.7, doc 19 section 1
Documents & Numbering base	Document header, lines, typed links, state history, attachments; document-type registry; numbering series with entity, location and till-position scopes; issuance protocol; gap detection	Doc 18 part C; addendum 24A after the tax advisor
Audit	Append-only audit store written through one facade in the business transaction; event-type catalogue; exception queues; hash chain in phase 2	Doc 18 part D, doc 19 section 4
Event backbone	Transactional outbox and archive, relay, broker publication, per-consumer inbox, replay	Doc 19 section 6
Edge sync gateway	Upload, distribution, heartbeat, series registration and counter transfer; rate limiting and backpressure	Doc 32
Internationalisation	ICU collation and formatting, message catalogue with English fallback, NFC normalisation, rendering pipelines; the input-method provisioning register per shop (configuration item)	Doc 19 section 5
Attachments	Object-storage keys and metadata, scoped like any record	Doc 18 (document_attachment)
Notifications delivery	Event-triggered sends through M9's adapter and templates, with retry	Doc 20 M9
Configuration register	Onboarding-time values (thresholds, lock windows, rounding, idle locks, resolution policies, offline account cap, quiet hours) with defaults, scoped per entity or location, audited	Doc 07 section 4; module designs section 7
Scheduling	End-of-day, expiry scans, statements, reconciliation, gap checks, ring progression; singleton under cluster locks	Doc 14 section 1.3

8. Event stream and queue

The event stream is the durable, replayable record of business facts; the work queue is a delivery mechanism for jobs. The design keeps them apart (doc 19 section 6 names the four kinds of event and their persistence).

8.1 Where events live

Decision (Proposed – ADR-30). The system of record for domain events is an append-only outbox archive in PostgreSQL written in the same transaction as the business change; a relay publishes rows to the broker; the broker holds nothing that cannot be re-derived from the archive. Central write volume is on the order of a thousand events a second at peak and a few thousand during reconnection bursts (doc 14 section 6), which PostgreSQL handles; replay and audit come free; there is one fewer stateful system; the broker is replaceable. The archive is partitioned by month and retained indefinitely.

8.2 Envelope

One envelope for every event, wherever it is created (doc 19 section 6.1): event id (UUIDv7), versioned dotted type, occurred and received times, source (central or device), dense per-source sequence, owning entity and location, aggregate type and id, correlation and causation ids, actor, and a small payload. The pair (source, source sequence) is the idempotency key for ingestion and the basis of gap detection.

8.3 Delivery guarantees and consumer rules

- **At-least-once delivery.** Every consumer is idempotent by event id; the kernel supplies a per-consumer inbox table.
- **Ordering per source, not globally.** Events from one till, or from central, apply in sequence; cross-source order is by received time with freshness stamps.
- **Consumers are registered in code**, reviewed, and listed in the module catalogue; the dependency graph stays visible.
- **Replay is first-class** for read models; layer 2 modules never replay: their state is corrected by compensating documents, never by re-processing history.
- **Poison events** go to a dead-letter queue with an alert after three attempts and never block a source's subsequent events.

8.4 Event catalogue by producer

Producer	Events	Main consumers (besides M8 and audit)
M1	entity.*, location.*, till_position.*, device.*, relationship.*, user.*, role.changed, external_grant.*, authentication.failed, account.locked_out	Numbering (series creation), edge sync (snapshot scope and permission snapshot), M4 (exposure configuration)
M2	sku.*, barcode.*, conversion.defined, tag.defined, tax_rate.published, image.*, batch.registered/corrected	Edge sync (catalogue snapshot), M3 (batch MRP review), M5 (alias mapping)
M3	price_list.published, discount_rule.activated/retired, control_price.entered/rescinded, mrp_policy.changed	Edge sync (pricing snapshot with apply_from), M4 (relationship price at acceptance)
M4	order.*, delivery_note.*, grn.captured/confirmed/closed, discrepancy.*, claim.*, invoice.issued, credit_note.issued, debit_note.issued, payment_receipt.recorded/disputed/reversed, cheque.cleared/bounced, exposure.updated, statement.generated, journal.postings_ready	M2 (batch registration), M5 (lots, receipts, returns), M7 (customer payments), M9 (journal export, e-invoice seam, notifications), M10 (purchase-order receipts)
M5	stock.moved, lot.negative, count.*, adjustment.approved, writeoff.*, repack.*, batch.expiring, opening_balance.recorded	M3 (in-stock MRPs for authoring checks), edge sync (lot snapshot), exception queues
M6 (via ingestion)	receipt.issued/voided/refunded, till_session.*, cash.dropped, drawer.opened_no_sale, mrp_cap.applied, mode.switched, sync.batch_received/anomaly, till.heartbeat	M5 (stock deduction per batch), M7 (account tenders), exception queues, fleet and sync health
M7	customer.*, account.*, banking.recorded	Edge sync (credit customer snapshot)
M8	report.rendered, export.ready, end_of_day.completed, projection.rebuilt, exception_report.generated	Notifications, named reviewers
M9	notification.sent/failed, journal.exported, api_client.registered	Audit

Producer	Events	Main consumers (besides M8 and audit)
M10 (P2)	purchase_order.issued, quotation.awarded, landed_cost.finalised, supplier_performance.computed	M5 (lot cost restatement), M4 (expected receipts)

8.5 Reference-data distribution to tills is not the event stream

Decision (Proposed – ADR-34). Tills do not subscribe to domain events. They receive versioned snapshots and deltas assembled per shop by the edge sync gateway: catalogue with thumbnails for the shop's assortment, prices, rules, control prices, tax, lot lists, credit customers and balances, permissions, message catalogue. Each snapshot carries a version and an apply_from business date; the till applies it atomically so a bill is never computed against a half-applied price list. Read-only at the till (P5).

8.6 Work queues

Queue	Behaviour
Edge ingestion	One job per uploaded batch; partitioned by device so one device's events apply in order, concurrent across devices; rate-limited per device at the gateway; duplicates acknowledged as already processed; documents and movements before audit rows within a device's window
Event relay	One queue per consumer so a slow consumer never delays another; dead-letter after three attempts
Rendering	A4 reports and statements to object storage; report.rendered on completion
Notifications	SMS and e-mail with retry and provider failover
Scheduled jobs	End-of-day aggregates, expiry scans, statement runs, reconciliation and gap checks, ring promotion, freshness recomputation; each singleton under a cluster lock

9. Edge architecture: the POS application

The Android application is the second place business logic runs. Doc 30 section 4 gives its shell and screens; doc 31 its update path; doc 32 its sync protocol. This section fixes its structure.

Component	Responsibility
Selling	Scan or key, weigh, resolve price through the shared engine (multi-MRP policy and control-price cap), rules, tenders as references, park and resume, receipt, same-session refund and void
Shop-management mode	Role-gated second mode on the same device hosting M4 and M5 screens: GRN capture and confirmation, count entry, write-off draft, shop repack, local reports. Entered park-first; entry and exit audited. Online-only functions open the web back office in the terminal's browser.
Shared engine library	The same library as central (ADR-31): price resolution, rule evaluation, tax, rounding, envelope, numbering counter. No server dependencies; versioned with the snapshot schema.
Local database	SQLite via Room, encrypted with a keystore-held key. Till-owned tables (documents, lines, state history, movements, audit, counters, outbox) and read-only cache tables replaced atomically per snapshot version. Synchronous commit on every issued document.
Local numbering	Persisted counters: the till-position series for receipts, voids, refunds; the location series for GRN, count, write-off draft and transfer when this device is the shop's primary till. Number and document commit in one local transaction; counter transfer is an audited procedure.
Sync worker	Uploads the outbox in sequence with acknowledgement cursors; downloads snapshot deltas; reports heartbeat, version and clock offset; never blocks the interface; passive status indicator
Peripheral layer	One interface per device class (scanner with GS1 parsing, printer, drawer, scale, customer display, label printer) with vendor adapters behind it and a simulator for development and training; UnifiedPOS categories as vocabulary
Receipt renderer	Raster output through the platform text stack with Noto fonts; also Z-report, day sales and session reports offline
Local authentication	Operator PIN against cached hashes; cached permission snapshot; idle lock; operator switch audited
Training mode	Separate local namespace with visibly non-live numbering and a watermark; uploads to the sandbox entity at central
Management agent (separate app)	Device-owner agent for provisioning, silent update, lock and wipe (doc 31); the POS application cooperates by reporting version and running self-tests

9.2 Offline-capable and online-only (H-02/H-03 as architecture)

Offline: fully local, uploaded later	Online-only: refused while disconnected, never queued
Cash sale, receipt, stock deduction, price and rule lookup, control-price check, park and resume, same-till same-session refund and void, GRN capture and confirmation, count entry, write-off draft, shop repack, account tender within the offline cap, Z-report and session report, operator authentication, training mode	New orders and lateral transfer requests, card, wallet and QR tender confirmation, cross-till or cross-day refunds, master-data changes, payment-receipt posting, write-off approval, any MFA-gated action. This resolves the FR-POS-050 wording conflict: blocked, not queued.

9.3 The channels between till and central

Channel	Direction	Carries	Owner
1 Event upload	Till to central	Issued documents with lines and state rows, movements, audit events, heartbeat	Edge sync gateway; doc 32
2 Reference distribution	Central to till	Versioned snapshots and deltas (8.5)	Edge sync gateway; doc 32
3 Software update	Update service to management agent	Signed packages, ring instructions, health reports	Doc 31
4 Online-only requests	Terminal browser to back-office API	Orders, statements, cross-session refunds, approvals	Back-office API; web shell

Channels 1 and 2 carry data in opposite directions and never overlap in what they carry (P5).

10. Integration surface

10.1 Inbound

API	Consumers	Design
Back-office API (public)	Back-office web; the accounting system; third parties	REST, OpenAPI 3.1, versioned under /v1; OIDC bearer tokens; scope and permissions enforced by the kernel and row-level security; idempotency keys on all mutating requests; cursor pagination; problem-detail errors. Every operation maps to a module command or query; no business logic in the API layer.
Edge sync API (private)	POS application	Device-credential authenticated plus operator context per batch; upload, acknowledge, fetch delta by version, heartbeat, series registration and counter transfer; rate-limited and backpressured; small, resumable responses.
Driver API (private)	Driver application	Delivery run, drop confirmation with signature, photo, timestamp, location; same envelope and idempotency.
Regulator and auditor access	Department of Co-operative Development; auditors	The back-office web with time-bound read-only roles and export; every grant and export audited.

10.2 Outbound adapters

Adapter	Phase	Design
Accounting export	P3 (mapping P1)	Consumes journal.postings_ready; produces postings in the target format once the package is chosen; mapping lives in M4, the adapter only translates.
Inland Revenue e-invoicing	P3 (readiness P1)	Adapter point on invoice.issued with the immutable snapshot; submission status is metadata; the invoice never changes.
SMS / e-mail	P1	Provider adapter behind kernel delivery; templates per language in M9; retry and failover.
Webhooks	P2/P3	Subscriptions to selected event types, entity-scoped, signed, retried; built on the same consumer mechanism.
Spreadsheet export	P1	M8; scoped and audited like any read.

10.3 Explicit non-integrations

Card, wallet and QR terminals (standalone; reference only; no PCI scope, ADR-19). Banks for payments or statements (ADR-01). Existing POS systems (none exist, R-01). Gazetted control prices (manual entry by a named Federation role; no feed exists). Google Play as an update channel is optional, not assumed (doc 31).

11. Data architecture summary

Choice	Statement
One database, schema per module (Proposed – ADR-29)	One PostgreSQL cluster; each module and kernel service owns a schema; no cross-schema foreign keys; references by identifier; a connection pooler in front.
Scope on every row	owner_entity_id (and location_id where bound) on every operational table; row-level security policy classes OWN, PARTY, FEDERATION_VIEW, EXTERNAL_TIMEBOXED (doc 18 section 3.7).
Documents are append-only	One base type in the kernel; no update statements for issued documents; state changes as history rows; corrections as typed links; application roles lack UPDATE and DELETE on document tables.
Ledgers, not balances	Stock, customer accounts and exposure are movement ledgers; balances are materialised and recomputable.
Outbox archive and audit trail	Two append-only kernel schemas, partitioned by month, retained indefinitely; audit written through one facade in the business transaction.
Partitioned volume tables	Documents, lines, movements, state history, outbox and audit range-partitioned by month from day one.
Text and collation	UTF-8, NFC-normalised at write; ICU collations on Sinhala, Tamil and English columns; trigram search; OpenSearch phase 2.
Binary content	Object storage for photographs, images, rendered documents, packages; keys and metadata in the database.
Read models	Rebuildable projections in M8's schema on the replica; never a source for transactional decisions.
Identifiers	UUIDv7 primary keys generated where the record is born; business numbers are separate unique attributes.

12. Deployment shape (provider-neutral)

Element	Minimum production shape on any provider
Application	At least two web/API and two worker instances of one container image behind a load balancer; stateless (P12); rolling deployment; an ingest-only worker role can be scaled during reconnection bursts.
Database	PostgreSQL primary with streaming replica; continuous write-ahead-log archiving to object storage for point-in-time recovery inside the 15-minute RPO; connection pooler; managed where available.
Broker and object storage	Persistent queues; S3-compatible store with versioning and a write-once bucket for audit anchors; MinIO on premises.
Identity, device management, update service	Self-hosted provider; device-owner MDM; update service with package store.
Environments	Development, test, staging (mirrors production; update ring 0), production. Training is a flagged sandbox entity inside production (ADR-35), not an environment.
Observability	Ingestion lag per till, series gaps, queue depth, replica lag, disk, ring status, fleet version distribution; the single DevOps engineer's eyes, part of the minimum shape.
Network	Inbound only through the edge and proxy over TLS; devices authenticated by credential; no inbound connections to shops.
Local hosting note	The shape runs on premises unchanged; what changes is who is on call at 02:00, the second site or cloud standby needed for the RTO, and the operating-model cost. Doc 14 section 5 carries the matrix.

13. Quality attributes

Requirement	Mechanism	Verified by
NFR-PRF-010/020 scale	Small write set at central; queued ingestion; read models on replica; partitioning; stateless scale-out	Spike 5 at the 15,000-till envelope
NFR-PRF-030 reconnection burst	Gateway rate limiting and backpressure; per-device ordered queues; idempotent acceptance	Burst simulation of two days of events from thousands of tills
NFR-PRF-040 15-minute poll	Versioned snapshot/delta channel	Delta size and apply time on the target terminal
NFR-AVL-010 RTO 4–8 h, RPO 15 min	Replica plus point-in-time recovery; restore runbook	Restore drill before pilot; quarterly
NFR-AVL-020/050 offline trading, GRN, refunds	Full local database, local counters, shared engine, one-writer split	Spike 1: 48-hour disconnected soak with power-loss injection

Requirement	Mechanism	Verified by
NFR-AVL-040 no scheduled downtime	Stateless instances; rolling deployment; expand-then-contract migrations	Deployment rehearsal under synthetic load
NFR-OFF-010/030 one writer; gapless offline numbering	P5 data split; persisted counters committed with the document	Ownership review of every edge table; counter tests under crash
NFR-OFF-040 future LAN peer visibility	Envelope carries source and sequence; a peer channel can reuse it (shop-hub option, doc 20 Appendix A)	Design review only (P2)
NFR-SEC-010/030 ASVS L2, credentials, TOTP	OIDC provider; kernel authorisation; MFA step-up; till PIN with lockout	ASVS checklist; penetration test before certification
NFR-SEC-020 no card data	No terminal integration; reference only	Data-model review: no card-number field exists
NFR-CMP-010 PDPA	Minimal PII in M7; scope enforcement; audited access; time-bound regulator role	PDPA review of doc 18
NFR-CMP-030 tax advisor review	Numbering isolated in one kernel service	Detailed design after the written review
NFR-USE-010..060 usability directive	Screens as paper; role-scoped screens; passive sync indicator; compensating documents; sandbox	Spike 7 cashier rounds
NFR-LOC-010/020 trilingual, ICU, thermal	Internationalisation kernel; translations as data; raster receipts	Spike 2 on target printers
NFR-UPD-010..050 updates; modular monolith	Device-owner agent, signed packages, rings, floor (doc 31); Modulith enforcement	Ring rehearsal staging to pilot; build fails on boundary violation
NFR-AUD-010/030 immutable documents; audit catalogue	Append-only schemas; privileges; single audit facade	Privilege test; coverage test against doc 07 section 3

14. Decisions proposed in this document (candidate ADRs)

ID	Decision	Rationale and consequence
ADR-28 (revised)	Nine phase-1 modules plus Procurement (P2) plus a kernel, four layers, rules R1–R7 enforced by build checks	Modular by mechanism; extraction possible; full-stack handover by module; phase-2 boundary exists from day one
ADR-29	One PostgreSQL cluster, one schema per module, no cross-schema foreign keys, row-level scope on every table	Single operational unit; module independence; scope at the data layer
ADR-30	Transactional outbox archive in PostgreSQL as the system of record for events; broker is delivery only	Replay and audit from the archive; one fewer stateful system; provider-neutral; consumers must be idempotent
ADR-31	One shared engine library (price resolution, rules, tax, rounding, envelope, counters) used by central and the Android application	Removes offline-bill-differs-from-online-bill defects; library must stay free of server dependencies
ADR-32	Spring Modulith (or ArchUnit) to declare and verify boundaries in the build	Enforcement must be automated
ADR-33	Provider-neutral infrastructure: containers, PostgreSQL, S3-compatible storage, mainstream broker, open-source proxy and observability	Hosting neutrality including a local option; some cloud conveniences deliberately unused
ADR-34	Reference-data distribution to tills as a versioned snapshot/delta channel, not domain-event subscription	Small stable till contract; atomic price-list application
ADR-35	Training is a flagged sandbox entity in production plus a till namespace with visibly non-live numbering	Real pipeline, no extra environment, no contamination
ADR-36	The POS application carries a role-gated shop-management mode; online-only shop functions use the web back office on the same terminal	Single-device shops; one package per shop
ADR-37 (new)	Trading documents (order to settlement) are one module, M4; the document base type and numbering are kernel	Ownership pivot and state machines in one place; one base for till and back-office documents
ADR-38 (new)	Batch identity in M2; stock lot and cost in M5; synthetic batches for non-batch-tracked SKUs	One physical batch, many successive owners at different costs; one price-resolution path
ADR-39 (new)	Stateless application tier: no sticky sessions, idempotency keys on mutations, cluster-locked singleton jobs, connection pooler	Add and remove instances at will; the database is the scaling ceiling by intent
ADR-40 (new)	Software updates through a device-owner management agent; no till-to-till package sharing in phase 1	Developer-verification exemption; single-till shops; NFR-UPD-040 change request

15. Risks, dependencies and open points

Item	Type	Effect and mitigation
Tax advisor review of numbering (ADR-11, NFR-CMP-030)	External dependency	Only the numbering kernel service changes; detailed design after the review; GRN fiscal-scope question added
Hosting decision (doc 14 DR-1)	External dependency	Affects operating model and achievable recovery objectives, not the module map; section 12 fixes the shape
NFR-UPD-040 LAN-shared updates	Change request needed	Deferred: the management agent downloads per device; package size discipline (under 30 MB) keeps this acceptable; revisit with the shop-hub option (doc 20 Appendix A)
Shared engine on Android (ADR-31)	Technical risk	Library kept lean; spike 1 runs identical bills at central and on the terminal against the same snapshot
Reconnection burst sizing	Technical risk	Backpressure and per-device queues; spike 5 measures and sets limits
Local database durability under power loss	Technical risk	Synchronous commit per issued document; battery-backed hardware; spike 1 injects power loss
Terminal selection (doc 14 DR-3)	Procurement dependency	Scale integration is the gating test; spikes 2, 3, 6 on candidates; M6 design starts against the minimum specification
Report catalogue workshop	Programme dependency	Pipeline designed now; definitions later; rebuildable read models make late additions safe
Control-price content ownership	Operational dependency	Named Federation owner required before pilot (doc 18 DR-8)
Catalogue data steward unnamed (T-03)	Programme risk	Blocks go-live, not design; M2 steward tooling designed early

Item	Type	Effect and mitigation
Identity provider product (doc 19 DR-1)	Design decision	Either candidate satisfies the architecture; the trade-off is operating footprint against feature completeness; decide with the broker in Stage 2 week 2
Doc 20 omission of Procurement	Documentation	Add M10 (P2) to doc 20 section 2 and the traceability summary (change request)

16. Next steps

- Design review of this version with the technical leads and the DevOps engineer; accept or amend ADR-28 to ADR-40; record outcomes in doc 09; raise the NFR-UPD-040 and doc 20 M10 change requests.
- Present sections 4, 6, 9.2 and 10.3 to Federation admins: the business-visible boundaries.
- Stand up the build skeleton: one repository, module descriptors for M1–M10 and the kernel, boundary verification tests, the document base, outbox and envelope in the kernel, the shared engine as a separate library module, two application instances in every environment from day one.
- Proceed with Stage 2 per doc 20: doc 21 (M1) and doc 32 (sync contract) this week; spikes 1 and 5 as soon as the envelope and outbox exist.

Appendix A. Glossary of abbreviations

Term	Meaning in this document
ADR	Architecture Decision Record (doc 09)
API / REST / OpenAPI	Application Programming Interface; the HTTP style used; its machine-readable description
ASVS	OWASP Application Security Verification Standard, Level 2
CAA	Consumer Affairs Authority of Sri Lanka; gazettes maximum retail prices
COOPFED	The national cooperative federation; the apex entity
EARS	Easy Approach to Requirements Syntax: the form used in the SRS
FEFO	First expired, first out
FR / NFR	Functional / non-functional requirement (docs 05, 06)
GRN	Goods received note; the moment ownership and risk transfer
GS1 Application Identifier	A prefix in a barcode identifying a data field (product number, batch, expiry)
ICU	International Components for Unicode
IRD	Inland Revenue Department
MDM / DPC	Mobile device management / device policy controller: the device-owner agent
MFA / TOTP	Multi-factor authentication / time-based one-time password
MPCS	Multi-purpose cooperative society
MRP	Maximum retail price printed on a pack; a batch attribute here
OIDC / OAuth2	OpenID Connect / OAuth 2.0
PCI	Payment card industry standards; avoided by never handling card data
PDPA / PII	Personal Data Protection Act No. 9 of 2022 / personally identifiable information
POS	Point of sale
RBAC	Role-based access control
RFC	Request for comments: a proposal circulated for stakeholder review
RLS	Row-level security: database policies filtering rows by the caller's scope
RTO / RPO	Recovery time / recovery point objective
SKU	Stock keeping unit
SRS	Software Requirements Specification (doc 05)
UUIDv7	Time-ordered universally unique identifier
VAT	Value added tax
Z-report	End-of-session till summary

End of document. v0.2 draft for design review; subject to revision after review and after the Stage 3 spikes report.